

# **SEVERE ACUTE MALNUTRITION**

Dr. Ali Hassen , MD

## Outline

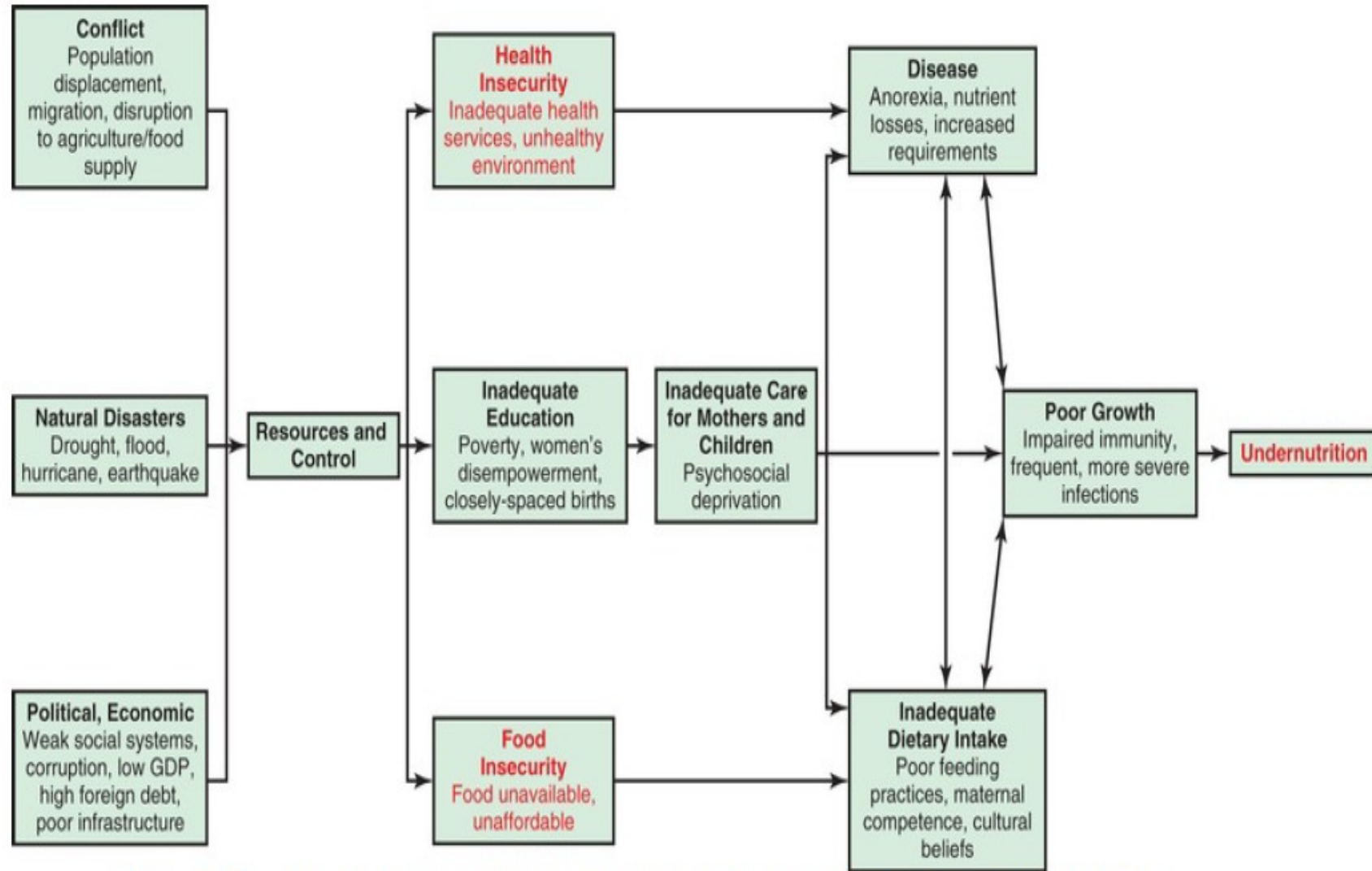
- Introduction
- Epidemiology
- Nutritional assessment
- Classification of undernutrition
- Patho-physiologic and adaptive changes
- Diagnosis and clinical features
- Management principles
- Dx & management of Complications

# Objectives

- To understand classification of malnutrition
- To understand magnitude of malnutrition nationally
- To know the different body system affected by severe acute malnutrition
- To understand the pathophysiology
- To know complications of SAM
- To understand management principle

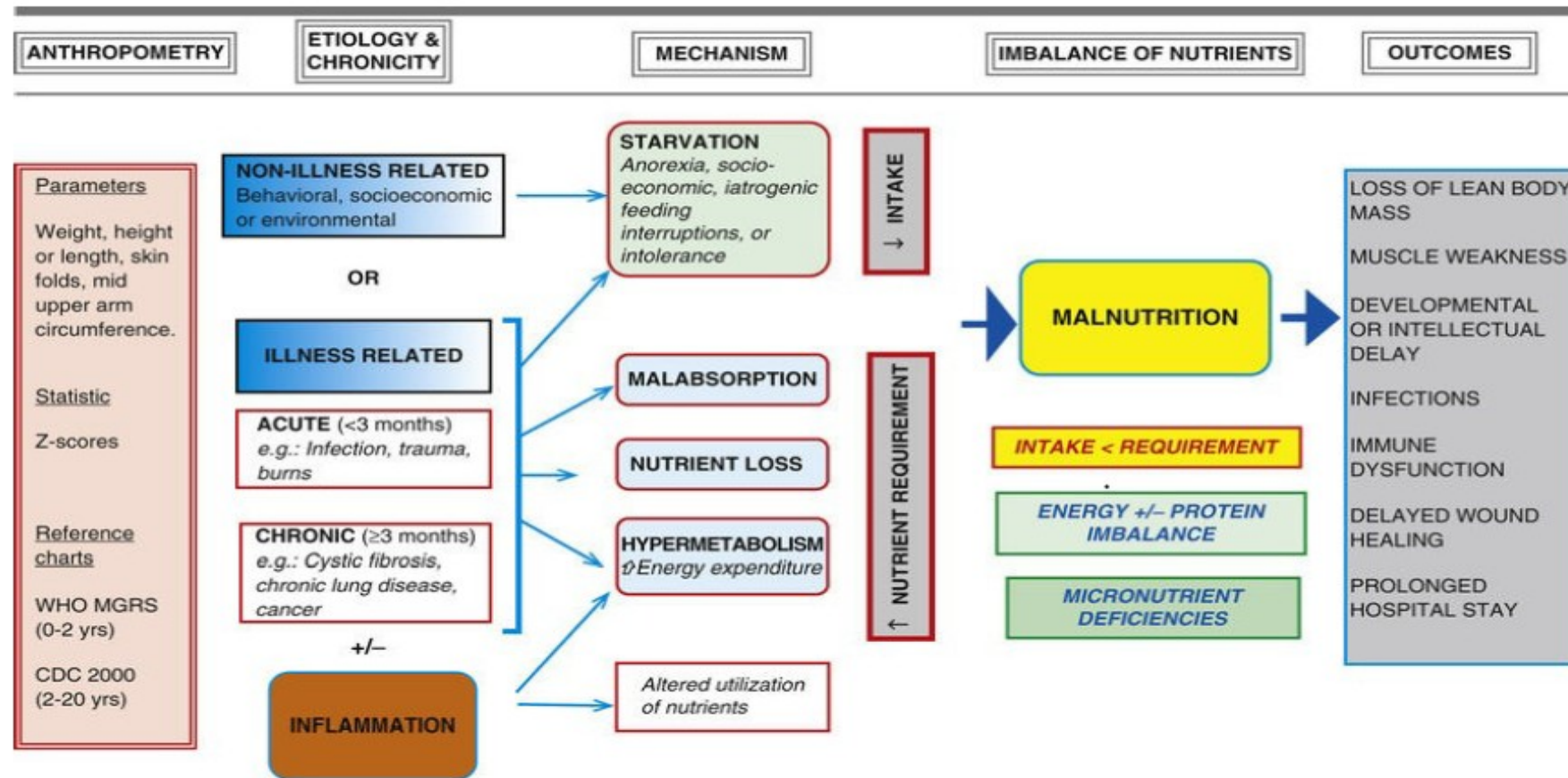
# Introduction

- Food insecurity indicates inadequate access to food for whatever reason.
- Hunger is the immediate physiologic manifestation of inadequate food intake.
- Undernutrition describes the biochemical &/or physical consequences of long-term inadequate intake.
- Food insecurity, hunger, and under-nutrition are viewed as a continuum, with food insecurity resulting in hunger, and ultimately, if sufficiently severe and/or of sufficient duration, in undernutrition.



**FIG. 57.1** Basic, underlying, and immediate causes of undernutrition.

# Etiology



**FIG. 59.1** Defining malnutrition in hospitalized children (ASPEN). CDC, Centers for Disease Control and Prevention; MGRS, Multicenter Growth Reference Study; WHO, World Health Organization. From Mehta NM, Corkins MR, Lyman B, et al: Defining pediatric malnutrition: A paradigm shift toward etiology-related definitions. JPEN J Parenter Enteral Nutr 37(4):460-481, 2013.

# Cont....

- The WHO defines malnutrition as the cellular imbalance between the supply of nutrients and energy and the body's demand for them to ensure growth, maintenance and specific functions.
- **PEM** – a condition that results when the body's needs for protein and energy fuels are not satisfied by diet also accompanied by deficiency of several micronutrients.

# Epidemiology

- In 2017, FAO estimated that about 821 million people, or 10.9% of the world's population, were undernourished, 98% of whom were in developing countries..
- Wasting (weight-for-height  $< -2$  SD) affects 7% of children  $<5$  year, with minimal change in prevalence over the past 2 decades.
- The global prevalence of stunting has declined from 33% in 2000 to 22% in 2017, with the greatest reductions occurring in Asia.
- Stunting prevalence is highest in the African region (30%).
- Infections can adversely affect nutritional status, and young children can quickly enter a cycle of repeated infections and ever-worsening malnutrition.
- For girls, the cycle of undernutrition is passed on to the next generation when undernourished women give birth to LBW babies.

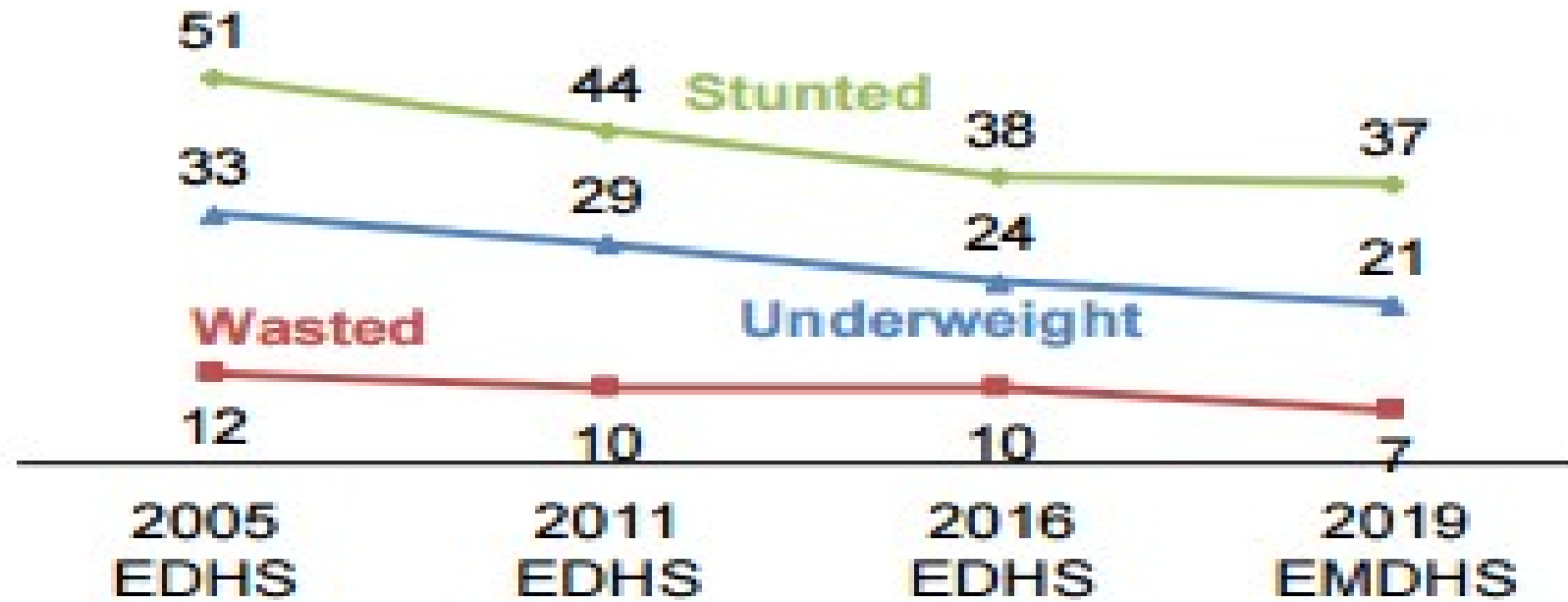
# Ethiopian Burden

## ➤ In Ethiopia according to EDMHS 2019:

- ✓ 37 % of children under age five are stunted (18% are severely stunted)
- ✓ Overall, 7% of Ethiopian children are wasted, and 1% are severely wasted.
- ✓ 21 % of children under age five are underweight.
- ✓ The proportion of underweight children is highest from 24-35 months (34 percent), which is the usual time for weaning.
- ✓ Somali and Afar having the highest percentages of children who are wasted, 21% and 14%, respectively.
- ✓ Children with severe acute malnutrition (SAM) have a nine-fold risk of mortality compared to well-nourished children hence the need for appropriate management

## Figure 5 Trends in nutritional status of children

*Percentage of children under age 5 who are malnourished*



# **Nutritional assessment**

- Medical and dietary history
- Anthropometric evaluation and physical examination
- Laboratory measurements

# Medical and dietary history

## ➤ **Medical history**

- Review of acute and chronic illnesses
- History of preexisting nutrient deficiencies
- Social history (poverty, domestic violence, parental employment, parental marital status, and parental substance abuse)
- Illnesses prior to or after development of malnutrition

## ➤ **Dietary history**

- Quantity and quality of current intake.
- Infant and young child feeding practices.
  - ✓ history of breast feeding pattern
  - ✓ formula preparation
  - ✓ volume consumed
  - ✓ feeding techniques

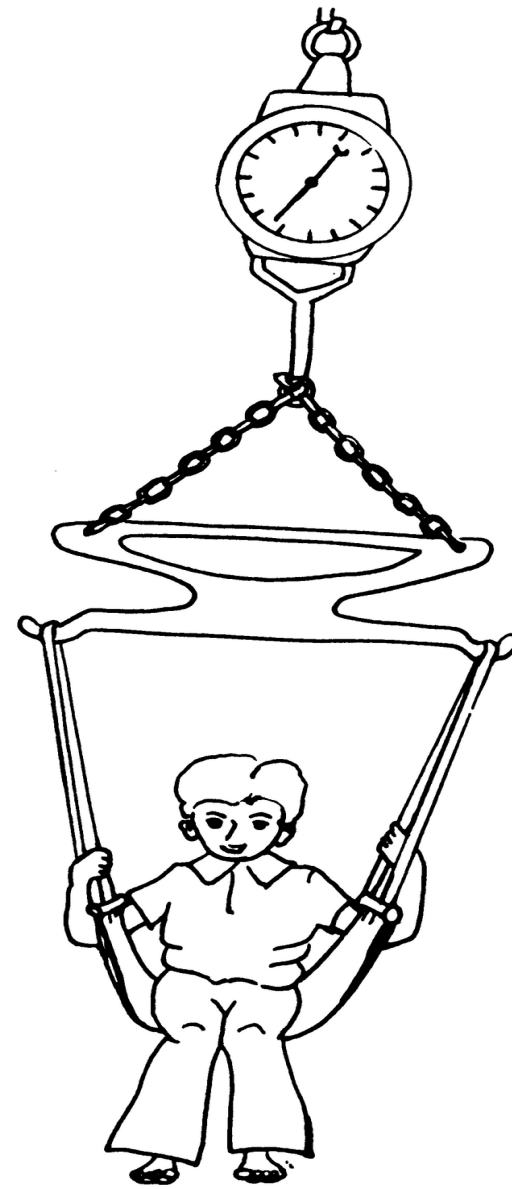
# Anthropometric Evaluation

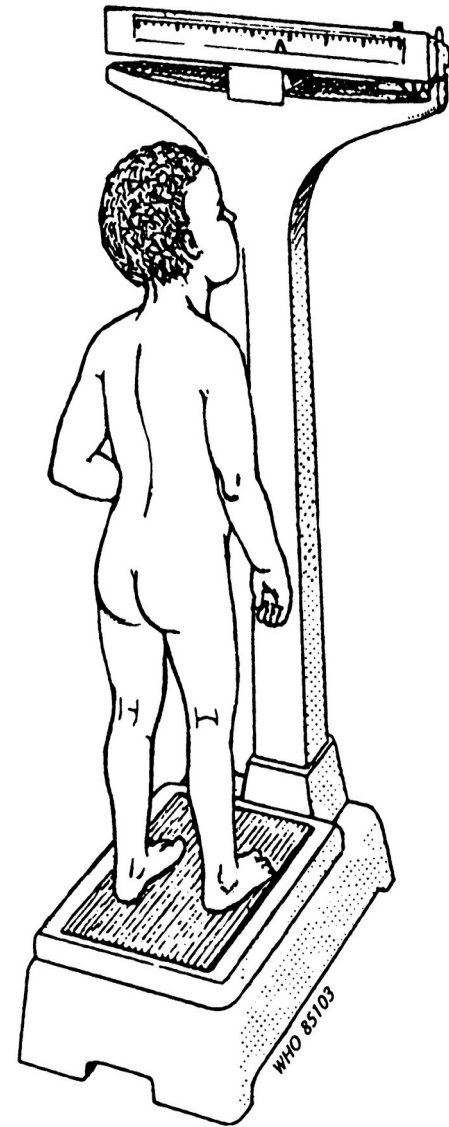
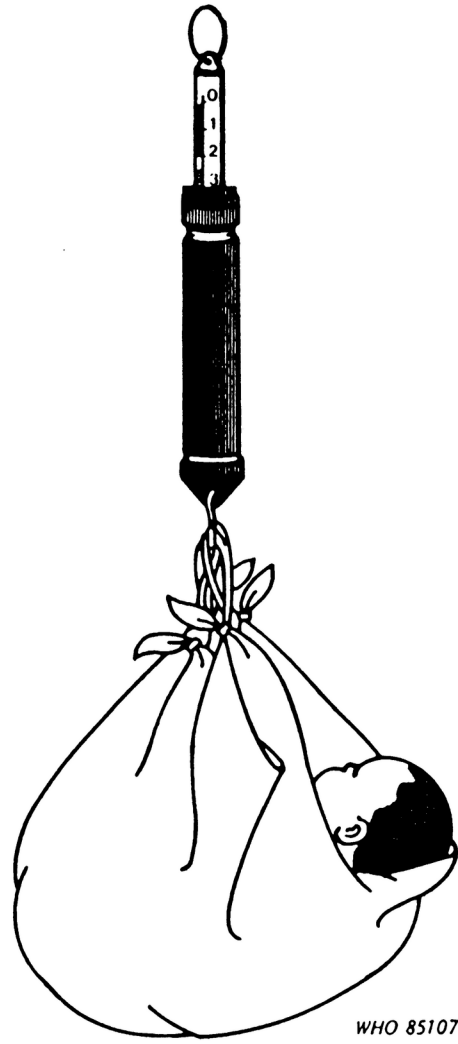
## Selection of measurements

- Age
- Weight
- Height/L
  - Length : supine for children aged less than 2 years
  - Height : standing for children aged 2 years or more
- MUAC:-6 months-59 months.
- BMI:-Above 5 years.

# Weight Measurement

- There are many types of scales currently in use.
  - Uni-scale (tared weighing)
  - Salter Scale (hanging scale)
  - Beam Scale
- For children below 6 years. A Salter spring balance with the scale measuring up to a maximum of 25 kg with calibration of 100 g.
- For children aged 6-10 years, a beam scale on which the child is made to stand..
- Both age groups readings are taken to the nearest 100 g.





# Height/Length measurement

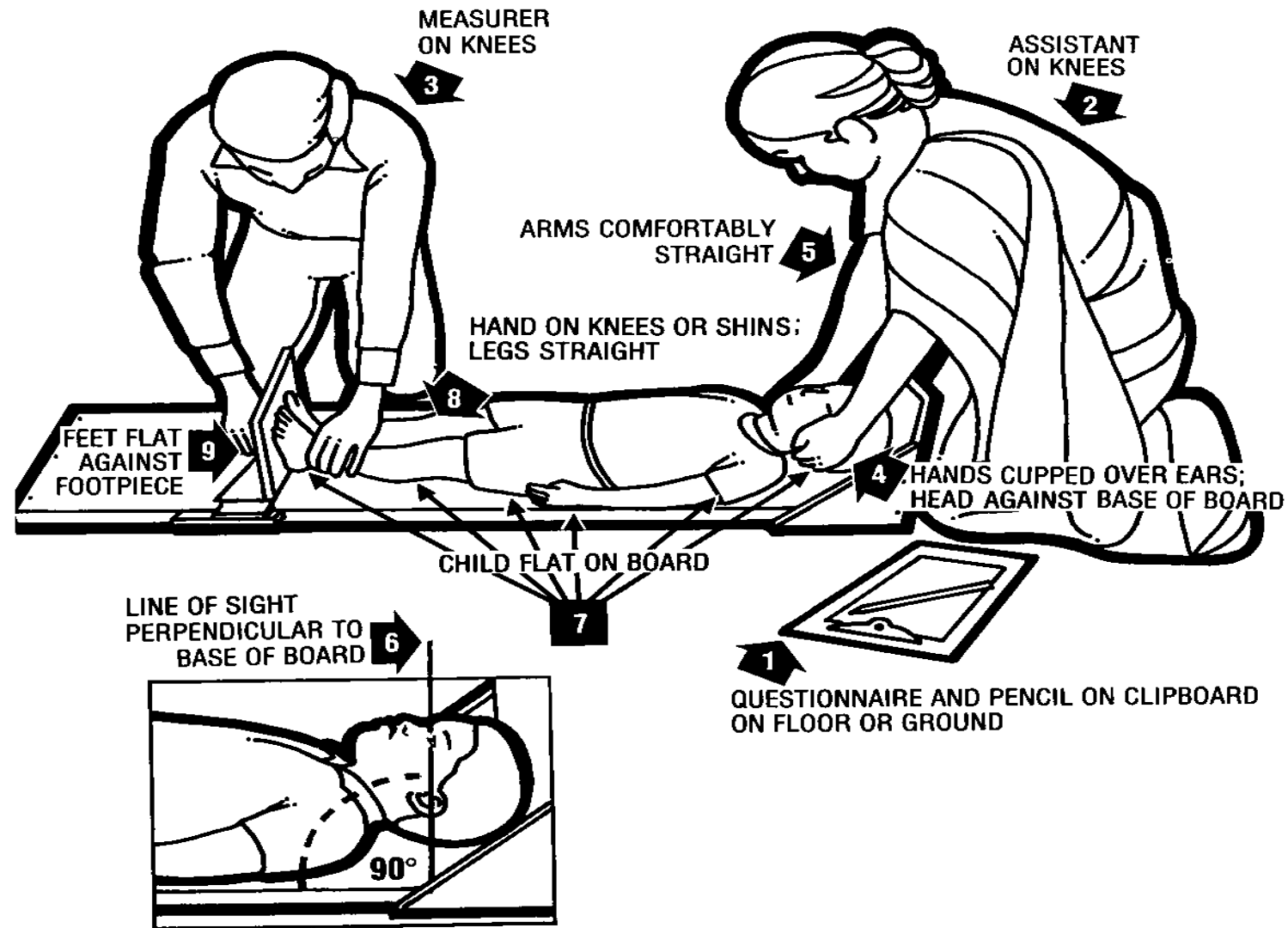
- Length is measured lying down on the back (recumbent).
- Height is measured standing upright.
  - For children under 2 years (or is less than 87cm or if the age is not available) measure the length.
  - For children 2 years or older (or is 87cm or more if the age is unknown) and able to stand measure the height.

**Note:** Length is usually greater than standing height by 0.7cm. If the child is 87 cm or more, but cannot be measured standing, subtract 0.7 cm from the supine length

## Cont.....

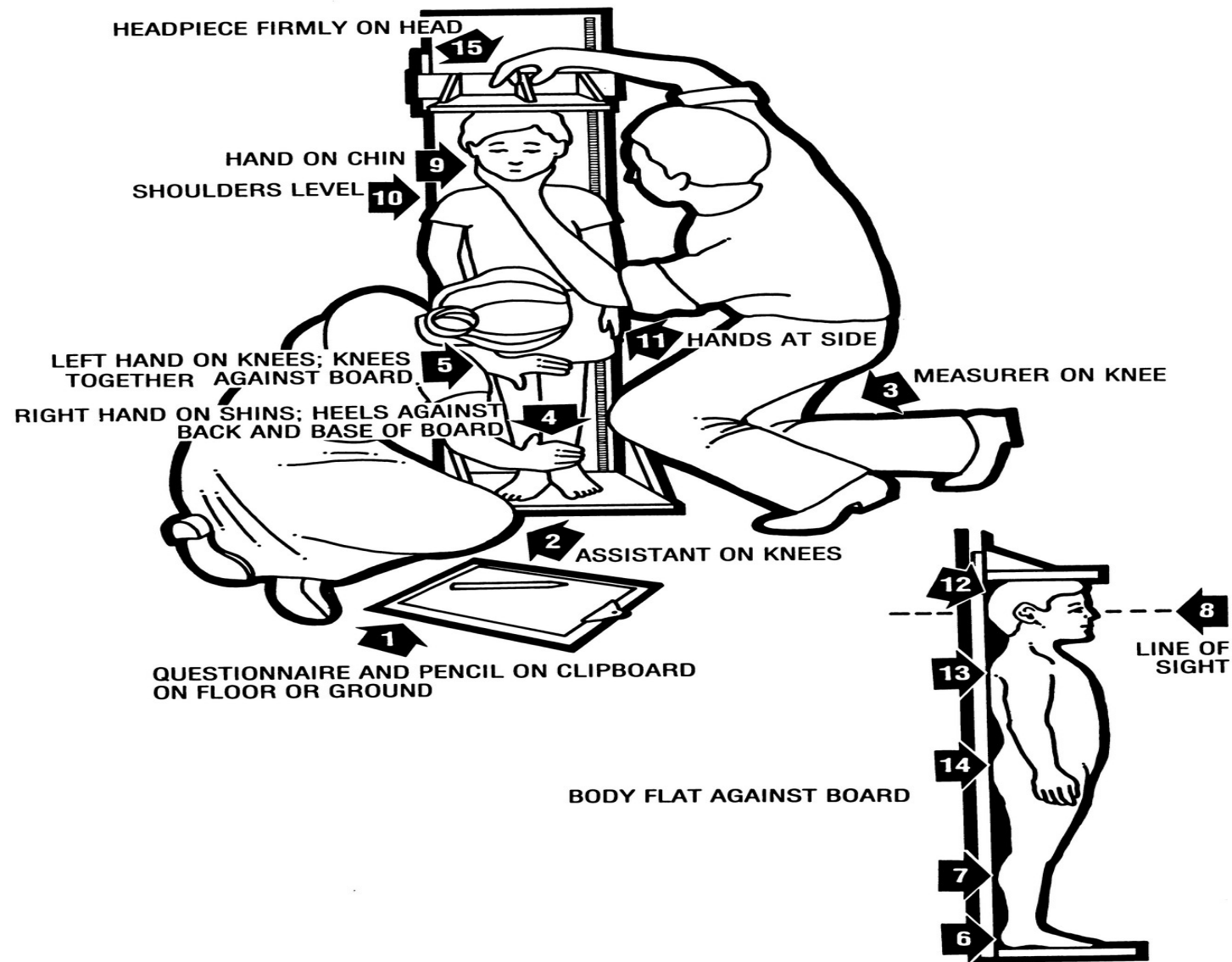
### ➤ **To measure length:**

- ✓ Infant is laid down on the board
- ✓ Head is positioned firmly against the fixed headboard with eyes looking vertically
- ✓ Knees are extended by firm pressure
- ✓ Feet are flexed at right angles to the lower legs
- ✓ The upright sliding foot-piece is moved to obtain firm contact with the heels and the length read to the nearest 0.1 cm



➤ **To measure Height:**

- ✓ Use a stadiometer with a vertical back board, a fixed base board, and a movable head board.
- ✓ The stadiometer should be placed on a levelled floor.
- ✓ Remove the child's socks and shoes for accurate measurement.
- ✓ Also remove hair ornaments and undo braids if they interfere with measurement.



# Indicators

- These three measurements are combined to form three indicators of nutritional status
  - Weight for age
  - Weight for height
  - Height for age

# Indices

- **W/A:** combined measurement:
  - NO individual diagnosis but trend assessment
  - For growth monitoring
- **W/H:** indicates degree of wasting
  - Individual diagnosis
  - Community diagnosis
  - Sensitive to change
- **H/A:** indicates linear growth retardation(stunting)
  - not sensitive to change
  - slow progress
  - Community diagnosis

# MUAC

## Advantages

- ✓ easy to perform
- ✓ "best" predictor of mortality in children

## Disadvantages:

- ✓ measurement error
- ✓ age dependency

## Classifications: 6 -59 months based on MUAC

- ✓  $< 11.5$  cm-----SAM
- ✓  $11.5 - < 12.5$  cm-----MAM
- ✓  $\geq 12.5$  cm-----No Malnutrition



# Cont...

## ➤ **Measure mid-upper arm circumference**

MUAC is measured on the upper left arm.

- To locate the correct point for measurement,
  - The child's elbow is flexed to  $90^\circ$ , with the palm facing upwards.
  - A measuring tape is used to find the midpoint between the end of the shoulder (acromion) and the tip of the elbow (olecranon) this point should be marked.
  - The arm is then allowed to hang freely, palm towards the thigh,
  - The measuring tape is placed snugly around the arm at the midpoint mark.
  - The tape should not be pulled too tight or too loose.
  - Read the measurement to the nearest 0.1 cm.

# Assessing and Classifying Acute Malnutrition in Infants 0-6 Months

| Signs                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Classify                                    | Treat              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------|
| <ul style="list-style-type: none"> <li>Any grade of bilateral pitting oedema (+, ++ or +++)</li> <li>OR</li> <li>WFL &lt; -3 z score</li> <li>OR;</li> <li>Recent weight loss or failure to gain weight.</li> <li>OR;</li> <li>Ineffective feeding (attachment, positioning and suckling) directly observed for 15-20, minutes, ideally in supervised separate area.</li> <li>OR;</li> <li>Presence of any of the following medical complications:</li> </ul> | SAM with and without medical complications. | Refer/Admit in SC. |

|                                                                                                                                                                                                                                                                         |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>WFH <math>\geq</math> -3 to &lt; -2 Z scores</li> <li>No bilateral pitting oedema</li> <li>AND</li> <li>No medical complications</li> <li>Clinically well and alert</li> <li><b>Mother has a MUAC of &lt; 23.0 cm</b></li> </ul> | MAM                    | <p>Assess the infant's feeding and counsel the mother or caregiver on appropriate IYCF practices. Emphasize on establishing effective exclusive breastfeeding.</p> <ul style="list-style-type: none"> <li>◆ If feeding problems, follow up in 5 days.</li> <li>◆ If no feeding problem, follow up in 30 days.</li> <li>◆ Admit the mother to TSFP.</li> <li>◆ Refer the mother for Productive Safety Net Programme (PSNP).</li> </ul> |
| <ul style="list-style-type: none"> <li>WFH <math>\geq</math> -2 Z score</li> <li>AND</li> <li>No bilateral pitting oedema</li> </ul>                                                                                                                                    | No acute malnutrition. | Congratulate and counsel the mother on appropriate IYCF practices.                                                                                                                                                                                                                                                                                                                                                                    |

## *Criteria for classification of acute malnutrition among children 6 to 59 months of age*

|                                                                                                                                                                                                                                                                               |                                   |                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>■ Bilateral pitting oedema + or ++<br/>OR</li> <li>■ MUAC &lt;11.5 cm<br/>OR</li> <li>■ WFH &lt;-3 z-score<br/>AND</li> <li>■ Appetite test passed</li> <li>■ No medical complications</li> <li>■ Clinically well and alert</li> </ul> | SAM without medical complications | Admit in OTP.<br><br>(See section 4.2 for the treatment and care plan)                                           |
| <ul style="list-style-type: none"> <li>■ MUAC <math>\geq 11.5</math> to &lt;12.5 cm</li> <li>■ WFH <math>\geq -3</math> to &lt;-2 z scores</li> <li>■ No bilateral pitting oedema<br/>AND</li> <li>■ No medical complications</li> <li>■ Clinically well and alert</li> </ul> | MAM                               | Admit in TSFP and counsel on appropriate IYCF practices<br><br>(See chapter 5.0 for the treatment and care plan) |
| <ul style="list-style-type: none"> <li>■ MUAC <math>\geq 12.5</math> cm</li> <li>■ WFH <math>\geq -2</math> z score<br/>AND</li> <li>■ No bilateral pitting oedema</li> </ul>                                                                                                 | No acute malnutrition.            | Congratulate and counsel the mother on appropriate IYCF practices.                                               |

# Classification

## 1)The origin of PEM can be

### ➤Primary malnutrition

- Resulted from inadequate food intake(food insecurity).
- ▶ Failure of lactation.
- ▶ Poverty.
- ▶ Cultural patterns.
- ▶ Lack of immunization & primary care.
- ▶ Lack of family planning.

## ➤ **Secondary malnutrition**

- Resulted from increased nutrient needs, decreased nutrient absorption, and/or increased nutrient losses.
- ▶ Infections.
- ▶ congenital diseases.
- ▶ Malabsorption.
- ▶ Metabolic causes.
- ▶ Psycho social deprivation.

# Classification...

## 2) CLINICAL PRESENTATION

- ✓ Edematous / **Kwashiorkor**
- ✓ Non edematous / **Marasmus**
- ✓ Marasmic-kwashiorkor

## 3) DURATION

- ✓ acute (Wt/Ht)
- ✓ chronic (Ht/age)

## 4) SEVERITY

- ✓ MAM
- ✓ SAM

# Recognize signs of acute malnutrition

- ✓ Visible severe wasting(marasmus) Cause  
Severe deficiency of both calories and protein
- ✓ Bilateral pitting edema(kwashiorkor) Mainly protein deficiency
- ✓ Dermatositis(+,++,++)
- ✓ Eye signs
- ✓ Flag sign
- ✓ And others

# Checking for Bilateral Pitting Oedema







# Flag sign



## **How does the physiology of severe acute malnutrition affect care of the child?**

### **➤ What is reductive adaptation?**

- ✓ The systems of the body begin to “shut down” with severe malnutrition.
- ✓ The systems slow down and do less in order to allow survival on limited calories. This slowing down is known as reductive adaptation.
- ✓ As the child is treated, the body’s systems must gradually “learn” to function fully again.
- ✓ Rapid changes (such as rapid feeding or fluids) would overwhelm the systems, so feeding must be slowly and cautiously increased.

- ❖ Decrease in energy intake → decrease in energy expenditure,
  - Metabolism of body fat → weight loss
- The poor availability of dietary proteins → Reduces protein synthesis loss of body protein(catabolism), primarily from skeletal muscle → Muscular wasting
- Adaptations lead to the
  - ✓ Sparing of essential body proteins &
  - ✓ Preserve essential protein dependent functions until the nonessential tissue proteins are depleted

**Why edematous SAM develops in some children and non-edematous SAM develops in others is unknown.**

Possible explanations

1. The variability among infants in nutrient requirements & in body composition at the time the dietary deficit is incurred.
2. Giving excess carbohydrate to a child with non-edematous SAM reverses the adaptive responses to low protein intake, resulting in mobilization of body protein stores. Eventually, Albumin synthesis decreases, resulting in hypoalbuminemia with edema. Fatty liver also develops secondary, perhaps, to lipogenesis from the excess carbohydrate intake and reduced apolipoprotein synthesis.

3. Aflatoxin poisoning as well as diarrhea, impaired renal function & decreased Na<sup>+</sup> K<sup>+</sup> ATPase activity.
4. Free radical damage has been proposed as an important factor in the development of edematous SAM. This proposal is supported by low plasma concentrations of methionine, a dietary precursor of cysteine, which is needed for synthesis of the major antioxidant factor, glutathione. This possibility also is supported by lower rates of glutathione synthesis in children with edematous compared with non-edematous SAM.

# End organ effects of malnutrition

- Body composition
  - Muscle and fat loss
  - Increased  $\text{Na}^+$
  - Decreased  $\text{K}^+$  and  $\text{Mg}^{++}$
  - Total body water increases (ECF)
- GIT
  - Villus atrophy
  - Production of gastric acid is reduced
  - Decreased intestinal motility
  - Pancrease is atrophied and reduced enzymes
  - Bacterial over growth

# End organ effects of malnutrition

- Liver
  - Reduced synthesis of protein
  - Impaired gluconeogenesis
  - Decreased metabolism of toxins and substances
- Cardiac
  - Myocardial atrophy, reduced CO, decreased BP
- Renal
  - GFR is reduced
  - Reduced excretion of excess water, Sodium and phosphate
- Hematology : anemia
- Metabolic
  - Hypoglycemia and Reduced metabolic rate by 30%.

# End organ effects of malnutrition

- Immunity
  - Impaired especially the Cell mediated
  - Reduced Ig A
  - Complement components are low
  - Reduced phagocytosis
  - Lymph glands, tonsils and the thymus are atrophied
- Inflammatory response
  - Impaired acute phase response
  - Reduced chemotaxis of WBC to site of infection/  
inflammation
  - Prone to infectious agents/subtle signs

# Cont...

- Endocrine system
  - Insulin levels are reduced and the child has glucose intolerance
  - IGF-1 levels are reduced, although growth hormone levels are increased
  - Cortisol levels are usually increased
  - T3 and T4 decreased

# Cont.....

- Skin, muscles and gland
  - The skin and subcutaneous fat are atrophied, which leads to loose folds of skin
  - Eyes may be sunken because of loss of subcutaneous fat in the orbit
  - Many glands are atrophied the child has dryness of the mouth and eyes and sweat production is reduced .
  - Atrophy of hair roots of the scalp

## Clinical Signs of Malnutrition

| SITE           | SIGNS                                                                                                                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Face           | Moon face (kwashiorkor), simian facies (marasmus)                                                                                                                                                               |
| Eye            | Dry eyes, pale conjunctiva, Bitot spots (vitamin A), periorbital edema                                                                                                                                          |
| Mouth          | Angular stomatitis, cheilitis, glossitis, spongy bleeding gums (vitamin C), parotid enlargement                                                                                                                 |
| Teeth          | Enamel mottling, delayed eruption                                                                                                                                                                               |
| Hair           | Dull, sparse, brittle hair; hypopigmentation; flag sign (alternating bands of light and normal color); broomstick eyelashes; alopecia                                                                           |
| Skin           | Loose and wrinkled (marasmus); shiny and edematous (kwashiorkor); dry, follicular hyperkeratosis; patchy hyper- and hypopigmentation ("crazy paving" or "flaky paint" dermatoses); erosions; poor wound healing |
| Nails          | Koilonychia; thin and soft nail plates, fissures, or ridges                                                                                                                                                     |
| Musculature    | Muscle wasting, particularly buttocks and thighs; Chvostek or Trousseau sign (hypocalcemia)                                                                                                                     |
| Skeletal       | Deformities, usually as a result of calcium, vitamin D, or vitamin C deficiencies                                                                                                                               |
| Abdomen        | Distended: hepatomegaly with fatty liver; ascites may be present                                                                                                                                                |
| Cardiovascular | Bradycardia, hypotension, reduced cardiac output, small vessel vasculopathy                                                                                                                                     |
| Neurologic     | Global developmental delay, loss of knee and ankle reflexes, impaired memory                                                                                                                                    |
| Hematologic    | Pallor, petechiae, bleeding diathesis                                                                                                                                                                           |
| Behavior       | Lethargic, apathetic, irritable on handling                                                                                                                                                                     |

From Grover Z, Ee LC: Protein energy malnutrition, *Pediatr Clin North Am* 56:1055–1068, 2009.

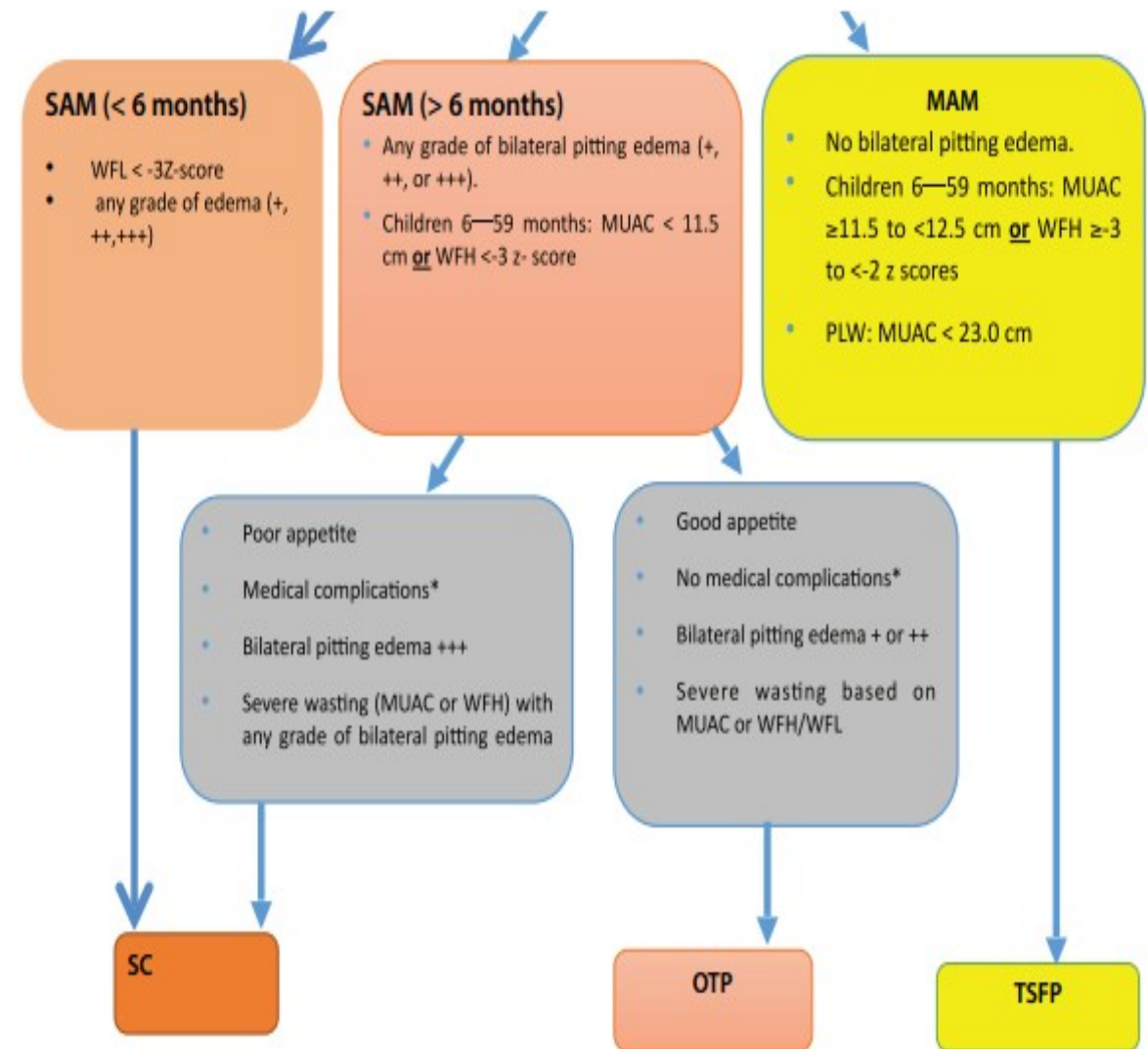
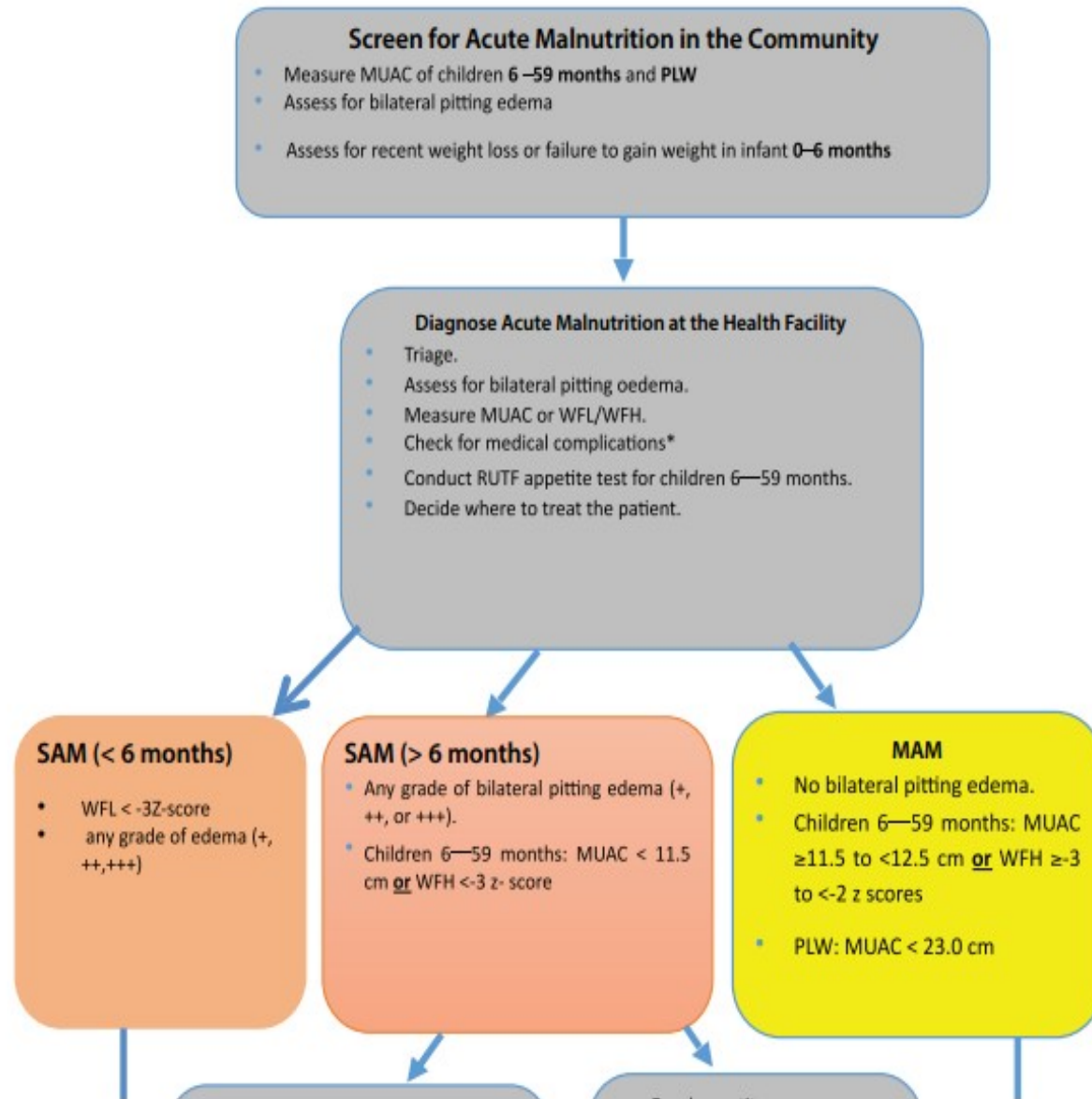
# Laboratory studies

- Blood glucose
- Hgb or HCT
- Blood film
- PIHCT
- Stool microscopy
- Urine analysis and culture
- Chest x-ray
- Tests for tuberculosis
- Serum concentrations of total proteins specially albumin
- Electrolytes

# Mgt of severe acute malnutrition

- Admission
- Inpatient mgt
  - Stabilization phase
  - Transition phase
  - Provide Sensory Stimulation and Emotional Support
- Discharge
  - Rehabilitation (catch up diet for rapid growth)
- Follow up

# Flow Chart on the Management of Acute Malnutrition



\*SAM medical complications include: Poor appetite, intractable vomiting, convulsions, lethargy, not alert, unconsciousness, high fever (axillary temperature ≥38.5 °C), lower respiratory tract infection (LRTI), dehydration, persistent diarrhoea, severe anaemia, hypoglycaemia, hypothermia (axillary temp. < 35 °C), severe skin lesions, eye signs of vitamin A deficiency.

\*\*MAM cases with medical complications need to be managed at TSFP and provided medical care based on the ICCM/IMNCI protocol.

# Assessment and classification for Admission

- All children who fulfil any of the following criteria have SAM and
- They should be admitted to a Therapeutic Feeding Program (SC or OTP).

## **Infants less than six months:**

- Weight –for- Length (WFL) < -3Z score OR
- Presence of any grade of bilateral pitting edema

## **Children 6 to 59 months:**

- Weight –for- Length (WFL) / WFH < -3Z score OR
- Presence of bilateral pitting edema OR
- MUAC <11.5 cm

# Cont...

- **In-patient care (SC):** All children with SAM need to be treated in an in-patient care facility if they fulfil any one of the following.
  - 1) Infants below six months of age with SAM OR
  - 2) Grade +++ edema OR
  - 3) Mixed type of malnutrition(marasmic - kwash) OR
  - 4) Children 6 to 59 months with SAM who have any one of the medical complications OR
  - 5) Children 6 to 59 months with SAM who have failed appetite test ORChildren with SAM and referred from OTP for in-patient care

# SC Entry category

- **New admissions** :-New case who meets the admission criteria for SC
- **Readmission**
  - Relapse:- Cured within the past 3 months and now meets the admission criteria for SC
  - Returned defaulters :-Defaulted within the past 3 weeks and has returned to continue treatment in SC.
- **Transfers-in** :-Referred to the SC because condition deteriorated in OTP or has moved in from another facility where s/he was receiving care in SC.

# Goals of management

- Resolving life-threatening complications
- Restoring nutritional status without disrupting homeostasis
- Ensuring nutritional rehabilitation

# Complications

How to diagnose and treat?

## ➤ **Treatment of complications in SAM**

1. Hypoglycemia
2. Hypothermia
3. Dehydration
4. Hypovolemic shock
5. Septic shock (infection)
6. Heart Failure
7. Severe anemia
8. Electrolyte imbalance
9. Micronutrient deficiencies
10. K. Dermatitis,
11. Corneal clouding and ulceration
12. Abdominal distension

# Treatment of Hypoglycemia

- Blood glucose  $<3$  mmol/L ( $<54$  mg/dl)
  - Important cause of death in 1<sup>st</sup> 2 days
  - Hepatic energy production from galactose and fructose is much slower
  - Gluconeogenesis is limited
  - Serious infection, not fed in the last 4-6hr, vomited or too weak to feed or waiting for admission
  - S/S include : hypothermia, limpness/drowsiness, lethargy, loss of consciousness, lid retraction
  - **Note:** -Sweating and pallor may not occur in malnourished children with hypoglycemia. Often the only sign before death is drowsiness.

## Treatment of hypoglycemia

- Give 50ml bolus of 10% glucose or 10% sucrose orally.
- If the child is alert, but not drinking, give the 50ml by NG tube.
- If child is lethargic, unconscious or convulsing, give 5ml/kg body weight of 10% glucose IV followed by NG tube.
- Start feeding F-75 half an hour after giving glucose and give it every half-hour during the first 2 hours. The amount being  $\frac{1}{4}$  of the 3 hourly amount.
- Recheck the RBS level after 2 hours and
  - If RBS is  $\geq 54$ mg/dl, change 3-hourly feeds (8 feeds per day) of F-75
  - If it is still low, make sure antibiotics and F-75 have been given. Keep giving F-75 every half-hour and treat with second line antibiotics.

# Prevent/treat Hypothermia

- Rectal temp  $<35.5^{\circ}\text{C}$  or axillary  $<35^{\circ}\text{C}$
- Children should always sleep with their caretakers, dry well after bath, keep away from open window, cover head
- **Feed F-75 every 2 hours, starting straight away.**
- Always give feeds throughout the day and night
- **Keep warm-** cover at night kangaroo technique, wrap with mother, hot drinks to mother, blanket
- Keep room temperature  $28-32^{\circ}\text{C}$
- **Treat hypoglycemia**
- Give second line antibiotics/treat infection

# Diagnosis and Treatment of DHN

- Misdiagnosis of dehydration & giving inappropriate treatment is the commonest cause of death in severe malnutrition
- All signs of dehydration in a normal child occur in a severely malnourished child who is not dehydrated
- Mainly based on history of significant fluid loss/diarrhea & recent change in child's appearance-sunken eyes proven by mother rather than PE
- Diagnosis is presumptive not definite

# Treatment of dehydration

- Never use the standard protocol for well nourished child & not ORS but RESOMAL
- Whenever possible hydrate only orally
- IV fluids are dangerous, use only if there is shock with loss of consciousness from confirmed dehydration
- Re-Hydrate the child until wt deficit is corrected (usually 5%)
- **Give 5 ml/kg Q30min for 1<sup>st</sup> 2 hours then 5-10 ml/kg/hour alternate hours with F75 for 10 hours.**

## DHN cont....

- ❑ If diarrhoea continues, give ReSoMal after each watery diarrhoea to replace stool losses and prevent dehydration:
    - < 2 years: give 50-100 ml after each watery stool
    - 2 years and older: give 100 – 200 ml after each watery stool.
  - ❑ All edematous children have high water and sodium retention hence over hydrated.
  - ❑ If there is definite watery diarrhea and child clinically deteriorating- replace with Resomal 30 ml per watery diarrhea.
- NB:-**In cases of profuse (acute) watery diarrhoea (e.g. cases of cholera), should be replaced by a standard ORS without changing the amounts and frequency

# ReSoMal

- An oral rehydration solution developed for use in severely malnourished children.
- It consists of the standard WHO rehydration oral solution (ORS) that has been modified by decreasing sodium and increasing potassium concentrations .
- This solution is used to correct the hyponatremia and potassium deficiency that occur in severe malnutrition.
- This solution also can be made by diluting one packet of the standard WHO-recommended ORS in 2liters of water and adding 50 g(4 tablespoons) of sucrose (25 g/L) and 40 mL (20 ml/L) of mineral mix solution.

## Treatment of Dehydration

ONLY Rehydrate until the weight deficit (measured or estimated) is corrected and then STOP- DO NOT give extra fluids to “prevent recurrence”

### Conscious

#### ReSoMal

Give:

- › 5 ml/kg every 30 minutes for the first 2 hours
- › 5-10 ml/kg once per hour, alternate with F-75 every 30 minutes for 10 hours

### Unconscious

#### IV Fluids

Ringer's Lactate Solution with 5% glucose OR 0.45% Saline solution with 5% glucose at 15ml/kg the first hour, then reassess.

- › If improving, give 15 ml/kg during second hour.
- › If conscious, give ReSoMal via NGT.
- › If not improving – assume other causes such as septic shocks

# Monitor the child who is taking ReSoMal

Monitor progress **every half hour for the first two hours and then monitor hourly**

## ➤ Signs to check

- Respiratory rate - Count for a full minute.
- Pulse rate - Count for 30 seconds and multiply by 2.
- Weight
- Urine frequency
- Liver size: mark before any infusion
- Stool or vomit frequency
- Signs of hydration - Is the child less lethargic or irritable?

# Cont...

## ➤ **Signs of improving hydration status**

- less thirsty
- less lethargic
- Slowing of rapid respiratory and pulse rates
- Passing urine
- Gaining weight with clinical improvement

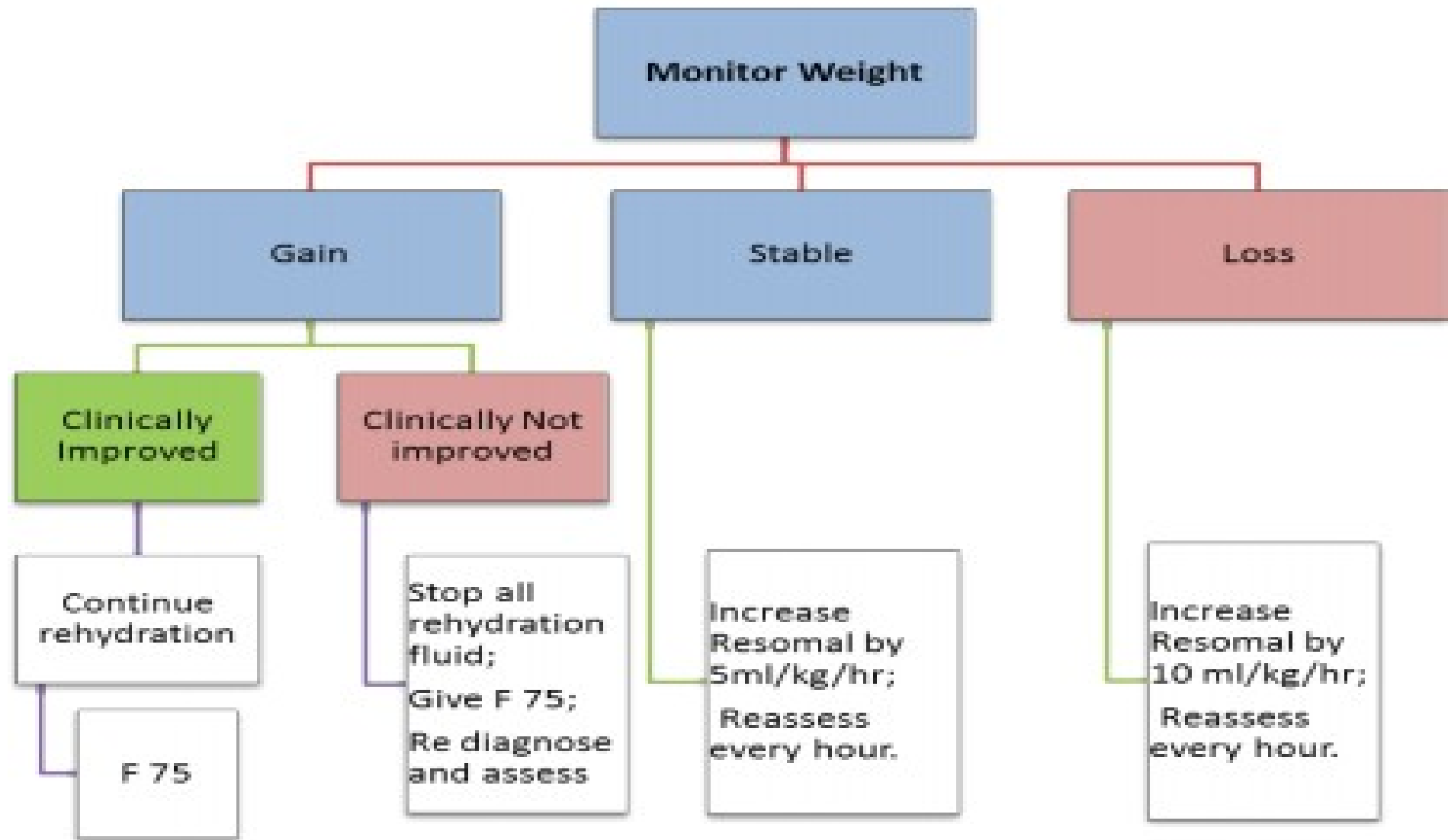
If a child has **three or more** of the above signs of improving hydration status, stop giving ReSoMal.

# Cont...

## ➤ **Signs of over hydration**

Stop ReSoMal if any of the following signs appear:

- Child's weight exceeds the target weight
- Increased respiratory rate by 5 breaths and pulse rate by 25 beats per minute.(Both must increase to consider it a problem.)
- Jugular veins engorged. (Pulse wave can be seen in the neck.)
- Sudden increase in liver size and tenderness
- Increasing oedema (e.g., puffy eyelids).
- Increasing weight with clinical deterioration



# Management of Shock in SAM patient

- The severely malnourished child is considered to have shock if he/she is lethargic or unconscious **and** has cold hands
  - plus either
  - ✓ slow capillary refill\* (longer than 3 seconds), or
  - ✓ weak, fast\*\* or absent radial or femoral pulses'

The four key steps to manage shock are

1. **Give oxygen**
2. **Give 10% Glucose 5 ml/kg by IV**
3. **Keep the child warm**
4. **Give IV fluids**

## Cont.....

- Use IV fluid-R/L with 5% dextrose or half strength(0.45%) NS with 5% dextrose
- Add 125 ml of 40% or 100 ml of 50 % glucose to 1 liter of 0.9% NS to make this fluid.
- If either of these are used, add sterile potassium chloride (20 Mmol/l) if possible
- Give 15 ml/kg over 1st hour & reassess, if still weight loss or stable weight repeat the same until weight gain but if no improvement assume septic shock and transfuse.
- As soon as child is conscious with good pulse-stop IV, give oral or by NGT
- Give 5-10 ml/kg ReSoMal per hour and continue to give F-75 alternate hours for up to 10 hours
- No IV drip near SAM patient who can drink or use NGT.

# Rx of septic shock

- Give maintenance IV fluids (4 ml/kg/hour) while waiting for blood.
- When blood is available, stop all oral intake and IV fluids,
- Give a diuretic(furosemide 1mg/kg) to make room for the blood, and then transfuse whole fresh blood at 10 ml/kg slowly over 3 hours.
- If there are signs of heart failure, give packed cells instead of whole blood as these have a smaller volume
- Give second & first line antibiotics together – IV
- Keep warm to prevent or treat hypothermia
- Give sugar water to prevent hypoglycemia – orally /NG tube
- If improved → F75

## Mgt of very Severe anemia

- If Hgb < 4 gm/dl or Hct <12%
- Give blood transfusion in the first 48 hours if:
  - Hgb is < 4 g/dl, (Hct is < 12 %), or
  - Hgb 4 to 6 g/dl (Hct 12 to 18%) and respiratory distress
- **Give Packed cell/whole blood 10ml/kg over 3 hrs.**
- Keep child NPO during and 3 hours after transfusion
- Do not transfuse a child b/n 48 hours and 14 days unless there is heart failure.
- **No Iron treatment in the first phase**

# Why no iron early in treatment ?

- Due to reductive adaptation, the severely malnourished child makes less Hgb than usual due to protein deficiency.
- Iron that is not used for making Hgb is put into storage.
- Thus, there is “extra” iron stored in the body
- So giving iron early in treatment will not cure anemia, as the child already has a supply of stored iron. Which leads to “free iron” in the body which can cause problems in **three** ways:
  - ❖ Free iron is highly reactive and promotes the formation of free radicals, which may engage in uncontrolled chemical reactions with damaging effects.
  - ❖ Free iron promotes bacterial growth and can make some infections worse.
  - ❖ The body tries to protect itself from free iron by converting it to ferritin. This conversion requires energy and amino acids and diverts these from other critical activities.
- Later, as the child recovers and begins to build new tissue and form more red blood cells, the iron in storage will be used and supplements will be needed.

# Anemia

(Check Hgb at admission  
if any clinical suspicion of  
anemia)

- Hgb  $\geq 4$  gm/dl, or
- Hct  $\geq 12\%$ , or
- Hgb 4 - 6 gm/dl (Hct 12 - 18%)  
AND no respiratory distress

- No need of transfusion
- Iron during phase 2

**If in the first 48 hrs of admission:**

- Hgb  $< 4$  gm/dl, or
- Hct  $< 12\%$ , or
- Hgb 4 - 6 gm/dl (Hct 12 - 18%) AND  
respiratory distress

- Give 10ml/kg whole or  
packed cells, 3hrs  
No food or fluid during  
transfusion

# Management of heart failure

## ➤ Diagnosis

- Physical deterioration(Increase in RR & PR) with weight gain
- Sudden increase in liver size
- Tenderness of the liver
- Crackles in lungs, grunting
- Prominent superficial and neck veins
- Engorgement of the neck veins when the abdomen (RUQ) is pressed,
- Increased or reappearance of edema, among other clinical signs and symptoms.

## ➤ Pneumonia is less likely in a child with RD after wt gain

# Treatment of heart failure

- Stop all oral or IV intakes/No fluid or food for 24-48 hours until managed
- Give sugar water to prevent hypoglycemia
- Furosemide 1 mg/kg single dose , repeat if necessary.
- If there is also severe anemia, treat the HF first than the anemia

# Give emergency eye care

## Eye signes

- Bitots' spots – superficial foamy white spots on the conjunctiva (white part of the eye).
- Pus and inflammation (redness) are signs of eye infection.
- Corneal clouding is seen as an opaque appearance of the cornea (the transparent layer that covers the pupil and iris).
- Corneal ulceration is a break in the surface of the cornea .If not treated, the lens of the eye may push out and cause blindness.
- Corneal ulceration is urgent and requires immediate treatment with vitamin A and atropine (to relax the eye)

## Give vitamin A

Vitamin A doses are given on **Day 1** and **Day 2, Day 15** if:

- The child has visible clinical signs of vitamin A deficiency
  - The child has signs of eye infection (pus, inflammation)
  - The child has measles now or has had measles in the past 3 months
  - The child has persistent diarrhoea
- You should instil one drop atropine (1%) into the affected eye(s) to relax the eye and prevent the lens from pushing out (for corneal clouding and corneal ulcer)
- Tetracycline eye drops and bandaging are also needed

Table 2: Dosage of Vitamin A

| Child's age   | Vitamin A Oral Dose |
|---------------|---------------------|
| < 6 months    | 50, 000 IU          |
| 6 - 11 months | 100, 000 IU         |
| ≥12 months    | 200, 000 IU         |

# Treat Infections

- Give all severely malnourished children antibiotics for presumed infection even if they do not have clinical sign of systemic infections.

*Table 13. Choice of Broad-Spectrum Antibiotics in SC*

| IF:                                                                                                                                                                             | GIVE:                                                                                                                                                                                                             |                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>NO MEDICAL COMPLICATIONS</b>                                                                                                                                                 | <b>Amoxicillin</b> oral: 25 mg/kg every 12 hours for 5 days or until referral to OTP                                                                                                                              |                                                                              |
| <b>MEDICAL COMPLICATIONS</b><br>(shock, hypoglycaemia, hypothermia, dermatosis with raw skin/fissures, respiratory or urinary tract infections, or lethargic/sickly appearance) | <b>*Gentamicin</b> IV or IM (5 mg/kg) once daily for 7 days, <b>plus:</b>                                                                                                                                         |                                                                              |
|                                                                                                                                                                                 | <b>Ampicillin</b> IV or IM (50 mg/kg) every 6 hours for 2 days                                                                                                                                                    | Followed by: <b>**Amoxicillin</b> oral (25 mg/kg), every 12 hours for 5 days |
| <b>Resistance to Amoxicillin and Ampicillin and Presence of Medical Complications</b>                                                                                           | In the case of <b>sepsis or septic shock</b> , give: IV/IM <b>ceftriaxone</b> (children or infants over 1 month of age (50 mg/kg every 8 to 12 hours) + oral <b>ciprofloxacin</b> (5 to 15 mg/kg 2 times per day) |                                                                              |
|                                                                                                                                                                                 | If <b>suspected staphylococcal infection</b> , add: IV/IM <b>cloxacillin</b> (12.5–50 mg/kg/dose four times a day, depending on the severity of the infection).                                                   |                                                                              |
| <b>Specific Infection Requires an Additional Antibiotic</b>                                                                                                                     | Add <b>specific antibiotic</b> as per the Ethiopia Standard Treatment Guidelines.                                                                                                                                 |                                                                              |
| <b>If HIV-positive or Child who is HIV Exposed.</b>                                                                                                                             | <b>Cotrimoxazole</b> oral according to the National Guidelines for Comprehensive HIV Prevention, Treatment and Care.                                                                                              |                                                                              |

\*If the child is not passing urine, gentamicin may accumulate in the body and cause deafness. Do not give the second dose until the child is passing urine.

\*\*If amoxicillin is not available, give ampicillin, 50 mg/kg orally every 6 hours for 5 days.

# Management of SAM in TB and HIV pt

- Management of SAM is the same
- Start nutritional treatment at least a week before ART, anti TB (except miliary TB) until the liver, pancreas and intestine have recovered sufficiently to metabolize the drugs safely
- If it is must to start in Phase I start with reduced dose
- Cotrimoxazole preventive therapy will not replace amoxicillin.
- Avoid toxic drugs
- Standard doses of drugs for HIV/TB are used in PII/OTP

# Management of Abdominal distension

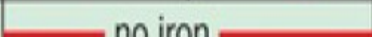
- Abdominal distension can occur because of reduced gastrointestinal motility due to electrolyte disturbance (Hypokalaemia).
- **The following measures should be taken:**
  - Give **Gentamicin** IV once daily plus **Ampicillin** IV every 6 hours
  - Consider adding Ceftriaxone IV
  - Stop all other drugs that may be causing toxicity (such as metronidazole)
  - Give a single IM injection of magnesium sulphate (2ml of 50% solution).
  - Pass an NG-tube and aspirate the contents of the stomach, then “irrigate” the stomach with 50 ml of isotonic clear fluid (5% dextrose or 10% sucrose –the solution does not need to be sterile). This should be repeated until the fluid that returns from the stomach is clear.

# Cont.....

- Put 5 ml/kg of sugar-water (10% sucrose solution) into the stomach and leave it there for one hour. Then aspirate the stomach and If the volume is less than the amount that was introduced, then either a further dose of sugar-water should be given or the fluid returned to the stomach.
  - There is frequently gastric and oesophageal candidiasis: give oral nystatin suspension or fluconazole.
  - Keep the child warm.
  - If the child's level of consciousness is poor give intravenous glucose,
  - Monitor the child carefully for 6 hours, without giving any other tt
  - Improvement is measured by a change in intestinal function and by improvement in the general condition of the child.

- **If there is intestinal improvement:**
  - Start to give small amounts of F-75 by NG tube (half the quantities)
- **If there is no improvement after 6 hours:**
  - Consider putting up an IV drip.
- Use Ringer-Lactate in 5% dextrose or half-strength saline in 5% dextrose. Kcl (20mmol/l) should be added to all solutions that do not contain potassium.
- The drip should be run VERY SLOWLY and should be not more than 2 to 4 ml/kg/h.
  - When the gastric aspirates decrease so that one half of the fluid given to the stomach is absorbed, discontinue the IV treatment and continue with oral treatment

# Cont .....

|                                            | Stabilization                                                                               |                                                                                     | Rehabilitation                                                                                |
|--------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|                                            | Day 1–2                                                                                     | Day 3–7                                                                             | Week 2–6                                                                                      |
| 1. Prevent/treat hypoglycemia              |          |                                                                                     |                                                                                               |
| 2. Prevent/treat hypothermia               |          |                                                                                     |                                                                                               |
| 3. Treat/prevent dehydration               |          |                                                                                     |                                                                                               |
| 4. Correct imbalance of electrolytes       |          |                                                                                     |                                                                                               |
| 5. Treat infections                        |          |                                                                                     |                                                                                               |
| 6. Correct deficiencies of micronutrients  | no iron  |                                                                                     | with iron  |
| 7. Start cautious feeding                  |          |  |                                                                                               |
| 8. Rebuild wasted tissue (catch-up growth) |                                                                                             |                                                                                     |            |
| 9. Provide loving care and play            |         |                                                                                     |                                                                                               |
| 10. Prepare for follow-up                  |                                                                                             |                                                                                     |          |

**FIG. 57.7** The 10 steps of treatment for severe acute malnutrition and their approximate time frames.

# The Do NOT DOs

1. Do NOT give DIURETICS to treat Edema
2. Do NOT give IRON during the Initial feeding phase
  - Add iron only after the child has been on F100 for 2 days
3. Do NOT provide high protein formula(over 1.5 gm/kg/day) in initial phase
4. Do NOT give IV Fluids routinely

# FEEDING

Cautious feeding is a critical part of managing SAM patients.

## **Stabilization Phase (phase I)**

- ✓ Required because of the patient's fragile physiological state and reduced homeostatic capacity.
- ✓ Life-threatening medical complications are treated
- ✓ We don't expect weight gain in this phase.
- ✓ Routine drugs are given to correct specific deficiencies
- ✓ Give small, frequent feeds of F-75(starter formula) every 2 hours, orally or by NGT including night.
- ✓ F-75 contains 75 kcal and 0.9 g protein per 100 ml.
- ✓ After the first day, increase the volume per feed gradually so that the child's system is not overwhelmed.
- ✓ If breastfed child, encourage the child to continue breastfeeding.

**Table 43-9 -- COMPOSITION OF F75 AND F100 DIETS**

| CONSTITUENT                | AMOUNT PER 100 mL              |                                 |
|----------------------------|--------------------------------|---------------------------------|
|                            | F75                            | F100                            |
| Energy                     | 75 kcal <sub>th</sub> (315 kJ) | 100 kcal <sub>th</sub> (420 kJ) |
| Protein                    | 0.9 g                          | 2.9 g                           |
| Lactose                    | 1.3 g                          | 4.2 g                           |
| Potassium                  | 3.6 mmol                       | 5.9 mmol                        |
| Sodium                     | 0.6 mmol                       | 1.9 mmol                        |
| Magnesium                  | 0.43 mmol                      | 0.73 mmol                       |
| Zinc                       | 2.0 mg                         | 2.3 mg                          |
| Copper                     | 0.25 mg                        | 0.25 mg                         |
| Percentage of energy from: |                                |                                 |
| Protein                    | 5%                             | 12%                             |
| Fat                        | 32%                            | 53%                             |
| Osmolarity                 | 333 mOsmol/L                   | 419 mOsmol/L                    |

*From World Health Organization: Management of severe malnutrition. Geneva, 1999, World Health Organization.*

# Feeding and NGT use

- Indications for NGT use
  1. Taking <80% of prescribed milk in 24 hrs in P-1
  2. Pneumonia with rapid RR/Respiratory distress
  3. Cleft palate or other oral deformities
  4. Painful mouth lesions
  5. Unconscious or lethargic child
- Use NGT only in P-I and for maximum of 3 days, try oral feeding at every feeding

# Cont...

- **Criteria for change of volume and frequency of F-75 feeds**
  - If vomiting, lots of diarrhoea, or poor appetite, continue 2-hourly feeds.
  - If little or no vomiting, modest diarrhoea (for example, less than 5 watery stools per day), and finishing most feeds, change to 3-hourly feeds.
  - After a day on 3-hourly feeds: If no vomiting, less diarrhoea, and finishing most feeds, change to 4-hourly feeds.

## **Criteria to move to transition phase**

1. Return of appetite(easily finishes F-75)
2. Reduced Edema
3. Clinically well
4. No NGT or IV line

# Transition Phase

- When the child is stabilized (usually after 2 to 7 days), the “catch-up” formula F-100/ RUTF is used to rebuild wasted tissues.
- F-100 contains more calories and protein: 100 kcal and 2.9 g protein per 100 ml.
- Transition can take around 2 to 3 days, during which RUTF is introduced alongside F-75.
- Increase the amount of RUTF over 2–3 days until the child takes the full requirement of RUTF.
- If the RUTF is not available or if the child does not accept it, give F-100 equal to the quantity of F-75 given in Phase 1
- Expected weight gain should be around 6 g/kg/day

## **Criteria to Move Back from the Transition Phase to the Stabilisation Phase**

- ✓ Weight gain of more than 10 g/kg/day in association with an↑ in RR
- ✓ Increasing or developing oedema
- ✓ A rapid increase in the size of the liver
- ✓ Any sign of fluid overload
- ✓ Tense abdominal distension
- ✓ A complication that necessitates an IV infusion
- ✓ A need for feeding by NGT
- ✓ Significant refeeding diarrhoea leading to weight loss

## **Criteria to Move from the Transition Phase to Rehabilitation**

- A good appetite: Passes the appetite test and takes entire amount of the daily RUTF ration or takes all amount of F-100 .
- Oedema reduced to moderate (++) or mild (+).
- If wasting with bilateral pitting oedema, oedema should completely disappear.
- Medical complications are resolving.
- Clinically well and alert.

# **Rehabilitation phase**

## **(Catch-up Diet for Rapid Growth)**

- Whenever patients have good appetite and no major medical complication they enter Phase 2
- Give RUTF (used in both in-patient & out-patient settings) or F100 (used in in-patient setting only) .
- Expected weight gain is at least 10g/kg/day.
- Patients who consume RUTF should be transferred from SC to OTP and monitored weekly at a facility close to his/her home
- If no possible referral for outpatient care, feed freely with F-100 to upper limit of 220 kcal/kg/day during rehabilitation
- Most children will consume at least 150 kcal/kg/day; any amount less than this indicates that the child is not being fed freely or is unwell..
- Start deworming and iron supplementation

## **Move back from rehabilitation phase stabilization phase**

1. Failure of the appetite test
2. Increase/development of edema
3. Develops any signs of a complication
4. Development of refeeding diarrhoea sufficient to lead to weight loss.
5. Fulfilling any of the criteria of “failure to respond to treatment”
6. Weight loss for 2 consecutive weighing
7. Static weight for 3 consecutive weighing

# Cont...

➤ **When do we say the child failed to respond**

- Failure to regain appetite after 4 days
- Failure to start to lose edema after 4 days
- Edema still present after 10 days
- Failure to gain at least 5 g/kg/day for 3 successive days after feeding freely on F-100

**Note:** The day of admission is counted as day 0

# CAUSES OF FAILURE TO RESPOND

- Problems with the treatment facility
- Problems of individual children

# Problems with the treatment facility

- **In-patients**

- Poor environment for malnourished children
- Failure to treat the children in a separate area
- Failure to complete the multichart correctly
- Insufficient staff (particularly at night)
- Poorly trained staff
- Inaccurate weighing machines
- Food prepared or given incorrectly

- **Out – patients**

- Inappropriate selection of patients to go directly to OTP
- Poorly conducted appetite test
- Inadequate instructions given to caretakers
- Wrong amounts of RUTF dispensed to children
- Excessive time b/n OTP distributions (e.g. two weekly gives significantly worse results than weekly visits)

# Problems of individual children

- **In –patients**
  - Insufficient food given
  - Food taken by siblings or caretaker
  - Sharing of caretaker’s food
  - Vitamin or mineral deficiency
  - Malabsorption
  - Psychological trauma (particularly in refugee situations and families living with HIV/AIDS)
  - Rumination
  - Infections
  - Other serious underlying disease:
    - Congenital abnormalities (e.g. Down’s syndrome),
    - Neurological damage (eg cerebral palsy),
    - Inborn errors of metabolism.
- **Out–patients** (In addition to all of the above)
  - Sharing within the family
  - Sibling rivalry (other children taking the diet)
  - All eating from the same plate (the malnourished child should always have his/her own portion of food).
  - Unwilling caretaker
  - Caretaker overwhelmed with other work and responsibilities

# The discharge criteria for children age 6-59 months

## ➤ **Discharged after Full Recovery (CURED) in SC:**

**Note:** The same anthropometric indicator that is used to identify and confirm SAM on admission should be used to determine recovery and discharge from treatment.

## ➤ **If admitted with bilateral pitting edema , discharge cured when**

- No bilateral pitting edema for 2 weeks
- AND MUAC  $\geq 12.5$  cm or WFH/WFL  $\geq -2$  z-score
- AND Clinically well and alert

## ➤ **If admitted with MUAC discharge as cured when**

- MUAC  $\geq 12.5$  cm
- AND No bilateral pitting edema
- AND Clinically well and alert

## ➤ **If admitted based on WFH/WFL, discharge cured when**

- WFH/WFL  $\geq -2$  z-score
- AND No bilateral pitting edema
- AND Clinically well and alert

# Feeding Infants less than 6 months

- Malnourished infants less than 6 months should always be treated in inpatient facilities.
- The management of complication is the same as children 6-59 months.
- The feeding of infants is different from older children and it is also different for infants on breast feeding or with a caretaker willing to breast feed and for infants who cannot be breast-fed.
- RUTF is not suitable for them, as the reflex of swallowing is not present yet.

# Cont.....

## ➤ **Diagnostic criteria:**

- Any grade of bilateral pitting edema (+, ++ or +++) OR
- WFL < -3 z score

## ➤ Severely malnourished young infants need:

- Treatment with antibiotics
- Warmth to treat and prevent hypothermia.
- Nutritional support for breastfeeding which may require supplemental suckling techniques.
- Continuous monitoring of weight and feed intake.
- Psychological support to the mother or caretaker and counselling on health and breastfeeding
- Keeping mothers and infants together must be a priority for a successful recovery of the infant

# Product used

## Diluted F- 100

### Why Should be diluted?

- Because babies of that age need more water and they are wasted, they need 100kcal/kg
- Should not be given undiluted F-100 at any time (owing to the high renal solute load and risk of hypernatremic dehydration).

Children less than 6 months, with oedema, should be started on F75 and not on F100diluted.  
When the oedema has resolved and they are suckling strongly they should be changed to F100  
diluted or infant formula

**Infants < six months with possibility of BF**

## ➤ **Stabilization**

- Breastfeed every 3 hours, at least for 20 minutes, more often if the child ask for more.
- One hour after breast-feeding, complete with F100 diluted using the supplementary suckling technique:
- Complete F-100 diluted: 130ml/kg/day (100kcal/kg/day),divided in 8 meals
- F-100-Diluted is prepared by adding 30% of the water, which was added to prepare the full strength F-100

# Supplemental Suckling Technique

- The mother holds the tube at the breast with one hand and uses the other for holding the beaker.
- The supplementation is given via an NGT n°8
- F-100 diluted is put in a beaker . The end of the tube is put in a cup.
- The tip of the tube is put on the breast at the nipple and the infant is offered the breast.
- When the infant sucks on the breast with the tube in is mouth, the milk from the cup is sucked up through the tube and taken by the infant.
- The beaker is placed 5-10cm below the level of the breast so the milk does not flow too quickly and distress the infant.



# Cont.....

## ➤ **Transition**

Monitor the progress of the infant in inpatient care by daily weighing.

- Decrease supplementary milk feeds by half when the child has gained at least 20g per day for 3 consecutive days and completely stop the milk when the child maintains a 10g per day weight gain and continue on breast milk alone
- If weight gain is not satisfactory when reducing the volume of milk supplement, increase the volume to the previous level for 2 days and try again.

- **Rehabilitation:** During this phase, the child does not receive any more milk supplement and is gaining weight by EBF. Observe feedings in order to ensure that the infant is feeding well, and as often BF must be at least 20 minutes. Continue to encourage the mother.

# Cont....

## ➤ **Discharge when the following criteria have been met:**

- Successful re-lactation and effective breastfeeding has been achieved.
- Gaining weight on exclusive breastfeeding (i.e., more than 5 g/kg/day for at least 3 successive days).
- No bilateral pitting oedema.
- Clinically well and alert.
- Infant has been checked for immunization and other routine interventions.
- Mothers or caregivers have been linked with community-based follow-up and support.

## **Infants <six months with no prospect to breast-fed**

### **➤ Stabilization phase**

- Appropriate replacement feeding should be given such as F-75 (for edematous) or diluted F-100 (for non-edematous) with relevant support to enable safe preparation and use.
- Give F75 or F-100 diluted using a cup and spoon base on child's weight
- If the infant is not taking sufficient amounts of milk orally, NGT can be used (drip, using gravity not pumping).
- NGT should not be used for more than 3 days and should only be used in the initial phase when the infant is still being treated for medical conditions and/or doesn't have appetite.
- Substantial weight gain is not expected during stabilization.

# Cont....

## ➤ **Transition phase**

- This phase should continue for 4-5 days
- When the infant shows the signs of recovery and is F-100 diluted:
  1. Increase the volume by 30%
  2. Monitor the infant's weight.
- When the infant shows the signs of recovery and is taking F-75(oedematous):
  1. Give F-100-Diluted provided at 150–170 mL/kg per day, or increased by one third over the initial amount given, providing 110–130 kcal/kg per day.
- The criteria for further increasing the amount of feed given to the child are as follows.
  - A good appetite: the child is taking at least 90% of the infant formula (F-100-Diluted) prescribed
  - Complete loss of bilateral pitting edema or
  - The child has been taking these amounts for at least 2 days
  - No other medical problem

# Cont....

## ➤ **Rehabilitation phase**

- During rehabilitation, infants should be fed using a cup and a saucer
- After 4-5 days, increase the volume of milk rations for another 30%.
- If the infant is still hungry after finishing the ration, give him more.  
Increase the rations of **5 ml per meal**.
- The infant's caregiver must prepare and provide meals under supervision while the infant is in the malnutrition ward.

# Cont....

## ➤ **Discharge criteria**

- Infant is feeding well with the replacement feed.
- Has adequate weight gain and has a WFL  $\geq -2$  z-score.
- No bilateral pitting edema .
- Clinically well and alert.
- Infant has been checked for immunization and other routine interventions.
- Mothers or caregivers have been linked with community-based follow-up and support

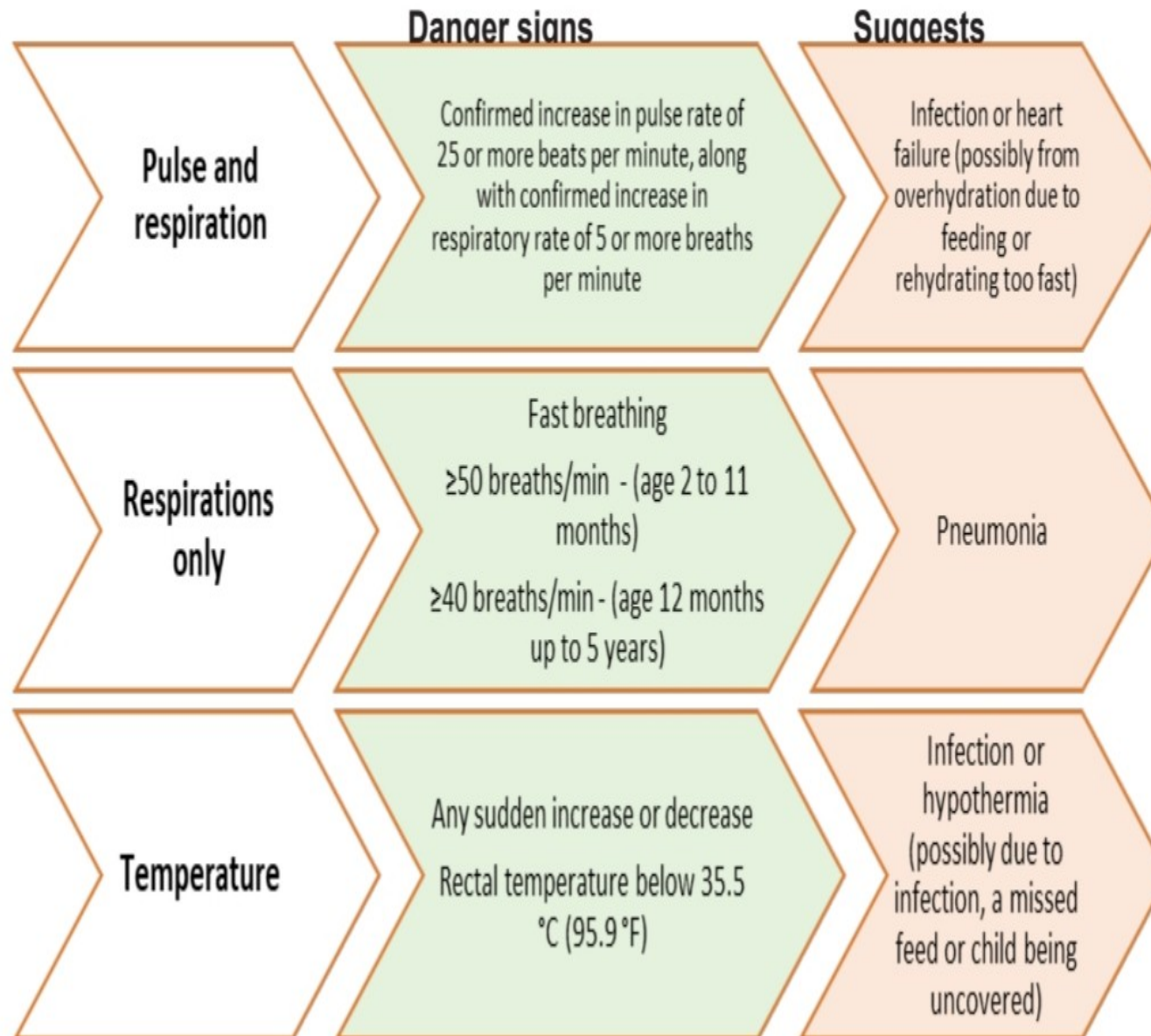
## SC Exit category

- **Cured:-**Has reached the discharge criteria for SAM treatment, i.e., the special cases that were not referred to OTP
- **Died :-**Dies while receiving treatment in the SC.
- **Defaulted:-** Absent for two consecutive days. Default should be confirmed.
- **Non-responder:-**Patient who remained in treatment in the SC does not reach the SAM discharge criteria after 16 weeks (4 months) in treatment.
- **Stabilised:-** Condition has stabilised and referred to continue treatment in OTP.
- **Transferred-out:-**Moved to another facility for further medical care *or* moved out to receive care in another SC

**Daily care**

- Handle the child gently
- Give Routine medicines and other medications
- Follow PR,RR,T and other danger signs
- Weigh the child daily
- **Eye care**
  - In addition to vitamin A Tetracycline, or Gentamycin eye drops are given for eye infection
  - Atropine eye drops are used to relax the eye when there is corneal involvement (i.e., corneal clouding or ulceration).In some cases, both types of eye drops may be needed.
    - Gentamycin (0.3%), (1%): 1 drop/ointment , 3-4 times daily for 7 to 10 days
    - Atropine (0.1%): 1 drop, 3 times daily for 3 to 5 days

## Summary of Danger Signs Related to Pulse, respirations and temperature



### ➤ Other danger signs include

- Anorexia (loss of appetite)
- Change in mental state (e.g., becomes lethargic)
- Jaundice (yellowish skin or eyes)
- Cyanosis (tongue/lips turning blue from lack of oxygen)
- Difficult breathing
- Difficulty feeding or waking (drowsy)
- Abdominal distention
- New edema
- Large weight changes
- Increased vomiting
- Petechiae (bruising)

# Cont...

## ➤ Skin care

- If the child has severe (+++) dermatosis, bathe for 10-15 min/day in 0.01% potassium permanganate solution.
- If potassium permanganate solution is not available gentian violet, zinc and castor oil ointment, petroleum jelly, or paraffin gauze dressing can be used.
- If the diaper area becomes colonized with Candida, use nystatin ointment or cream after bathing.
- Leave off diapers (nappies) so the affected area can dry. Be sure to dry the child well after a bath and wrap the child warmly.

## ➤ **Discharge Outcomes**

1. Cured:- Has reached the discharge criteria for SAM treatment
2. Died:- Dies while receiving treatment in the SC.
3. Defaulted:- Absent for two consecutive days in the SC
4. Non-responder:- Patient who remained in treatment in the SC does not reach the SAM discharge criteria after 16 weeks (4 months) in treatment.
5. Stabilized;- Condition has stabilized and referred to continue treatment in OTP.
6. Transferred-out :- Moved to another facility for further medical care or moved out to receive care in another SC

# Involving Mothers in child care

- There are many ways to encourage mothers' involvement in hospital care.
- Mothers can be taught to
  - Prepare food
  - Feed children
  - Bathe and change children and
  - Play with children, supervise play sessions, and make toys for emotional and physical stimulation.

# **OUT-PATIENT THERAPEUTIC PROGRAM (OTP)**

## **Admission procedure to OTP**

Step 1: Triage

Step 2: Assess for nutritional status

Step 3: Clinical assessment

Step 4: Perform the appetite test with RUTF

Step 5: Counselling in OTP and use of RUTF

Step 6: Dispense medication and RUTF

Step 7: Follow Up in OTP

## Criteria for OTP (6 month-59 months)

All children with SAM, which means

- Weight for height  $< -3$  sd
- MUAC  $< 11.5$  cm
- Edema (+/++) and:
  - Who passed appetite test (If the child eats some of the RUTF within 30 minutes )
  - Clinically well
  - Alert
  - No +++ edema
  - Do not have marasmic-kwashiorkor

## ➤ **Appetite test**

– Why do we do appetite test?

- No signs of clinical infection in children with SAM
- No perfect correlation between metabolic malnutrition & anthropometric malnutrition
- Metabolic malnutrition is the common cause of death and often its sign is loss of appetite
- Appetite test is the most important criteria to admit SAM either to inpatient or OTP
- Poor appetite means significant infection or metabolic dysfunctions like liver dysfunction, electrolyte imbalance and high risk of death.

➤ **OTP entry category**

- **New admission** :New case who meets the admission criteria for OTP
- **Readmission.**
  - Relapse**: Cured within the past 3 months and now meets the admission criteria for OTP.
  - Returned defaulter**: Defaulted within the past 3 months and has returned to OTP
- **Transfer-in** Continuing treatment in OTP after stabilisation in SC *or* has moved in from another OTP facility.

**Table 12: Routine Medication and Treatment in OTP**

|                                                              |                                                                          |             |                                                                           |                                                   |
|--------------------------------------------------------------|--------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------|---------------------------------------------------|
| 10% (2 tea spoon sugar in 100 ml water) sugar water solution | During referral to SC to prevent hypoglycaemia                           | All ages    | 50 ml                                                                     | Once, orally                                      |
| Amoxicillin                                                  | On admission of all SAM patients                                         | All ages    | 25 mg/kg, every 12 hours, for 5 days.                                     | Oral: 25 mg/kg, every 12 hours, for 5 days        |
| Albendazole<br><i>Or</i><br>Mebendazole                      | At the second visit<br><br>for treatment in OTP                          | ≥ 24 months | Albendazole 400 mg<br><br>Mebendazole 500 mg                              | Single dose, orally                               |
| Measles vaccine                                              | On the fourth week if the child has not yet received the measles vaccine | 9–59 months | Refer to the National Expanded Programme on Immunisation (EPI) Guidelines | Refer to the National EPI, ICCM/ IMNCI Guidelines |
| Other vaccines                                               | Update vaccinations based on EPI schedule                                | All ages    | Refer to the National EPI Guidelines                                      | Refer to the National EPI, ICCM/ IMNCI Guidelines |

WHO. 2010. *Guidelines for the Management of Severe Acute Malnutrition*. Geneva: World Health Organization.

# Exit category for OTP

- **Cured:** Has reached the discharge criteria for SAM treatment.
- **Died:** Dies while receiving treatment in the OTP.
- **Defaulted:** Absent for two consecutive visits. Default should be confirmed.
- **Non-responder:** Does not reach the SAM discharge criteria after 16weeks (4 months) in treatment.
- **Transfer out:** Condition has deteriorated or not responding to treatment according to action protocol and referred for treatment in the SC or moved out to receive OTP in another facility.

# MANAGEMENT OF MODERATE ACUTE MALNUTRITION (MAM)

- MAM is defined by a MUAC  $\geq 11.5$  cm and  $< 12.5$  cm or a WFH  $\geq -3$  z-score and  $< -2$  z-score (WHO standards) in children 6-59 months old.
- PLW with MUAC  $< 23$  cm.
- SFPs aim to prevent and rehabilitate acute malnutrition.
- Provide a two-week supply of specialised nutritious foods
- Fortified blended food (FBF) e.g., Super Cereal Plus, and RUSF(1 sachet daily ) are the two types of specialised nutritious foods used for management of MAM in Ethiopia.
- Follow-up visits conducted **every two weeks**.
- Check and provide routine medications such as vitamin A, deworming, and routine immunizations

# Types of SFP

## ➤ **Blanket Supplementary Feeding Programme (BSFP)**

BSFP is implemented when the Global Acute Malnutrition (GAM) levels are high and food insecurity is severe as part of an emergency response or as a response to seasonal shocks

- The admission and discharge criteria for BSFP does not rely on anthropometric indicators.
- Define the targeted groups, for example, all children aged 6-23 months or 6-59 months, or all PLW.
- BSFP normally operates for a period of about 3-6 months, after the situation has stabilized, it is closed and those registered are discharged

➤ Screen the individuals by measuring MUAC and assessing for bilateral pitting oedema. The screening is conducted to identify cases who may require referral for SAM management. It is not used for admission to the BSFP.

➤ **Targeted Supplementary Feeding program (TSFP)**

❑ Are generally used for treatment of MAM within individuals based on anthropometric admission criteria..

A. Children 6-59 months :-MUAC >11.5 to <12.5 cm OR WFH >-3 to <-2 Z scores AND No bilateral pitting oedema Clinically well and alert

B. PLW :MUAC < 23.0 cm

# References

- WHO guideline updates on the management of severe acute malnutrition 2013
- Guidelines for the Management of Acute Malnutrition Government of Ethiopia Federal ministry of health , April 2019
- Nelson text book 21<sup>st</sup> edition
- Modern nutrition in Health and Disease, 10<sup>th</sup> edition
- The 2019 Mini Ethiopia Demographic and Health Survey(EDHS)

**GALATOOMAA !**